R for Health Data Research

Beginner, Intermediate and Advanced

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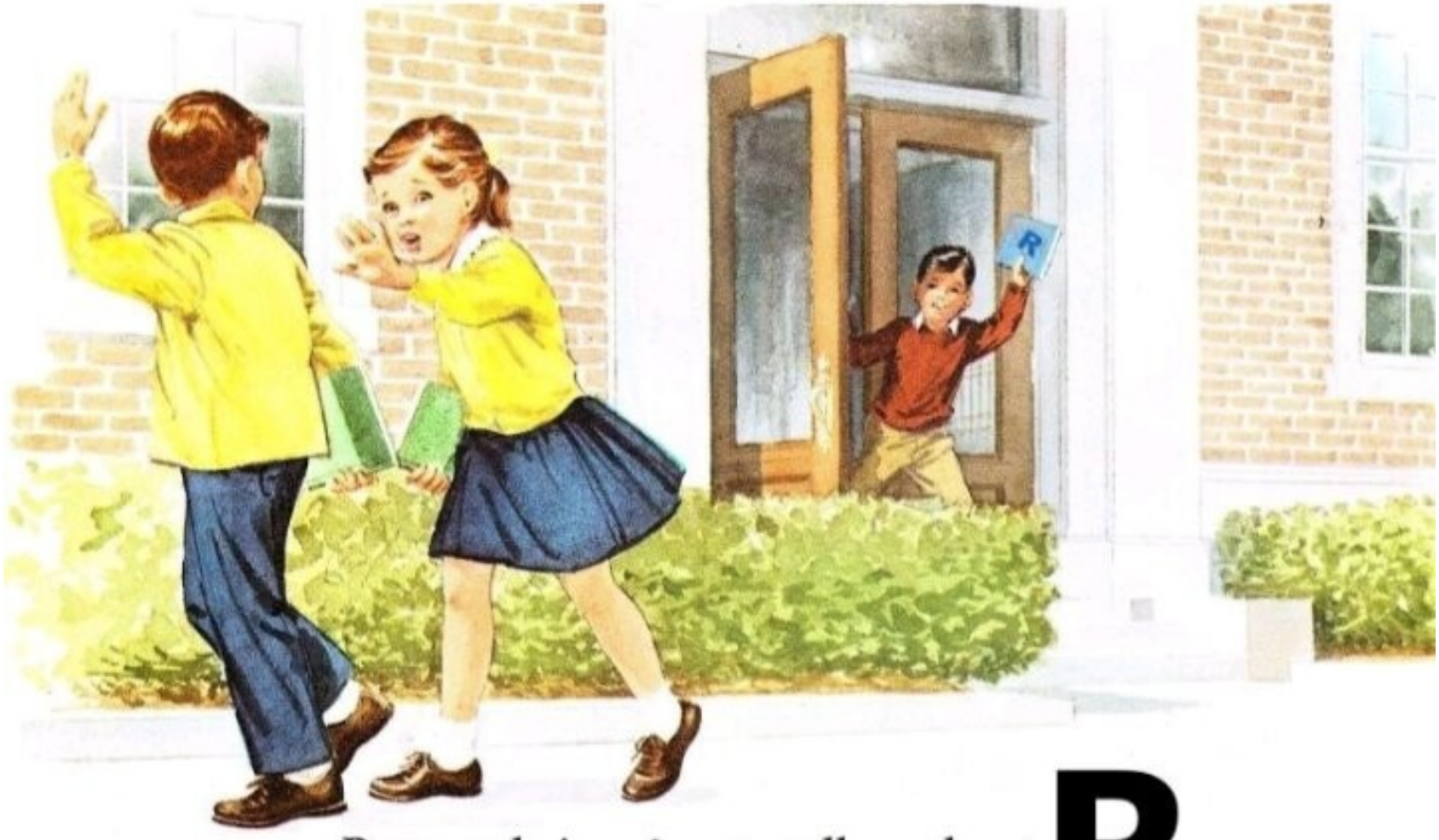


Methods

Clinical prediction models
Causal inference
Longitudinal modelling
Machine learning and AI

Applications

Mental health
Mental and physical health
Electronic health records
Digital interventions



Run, or he's going to tell us about **R** again!

Course materials

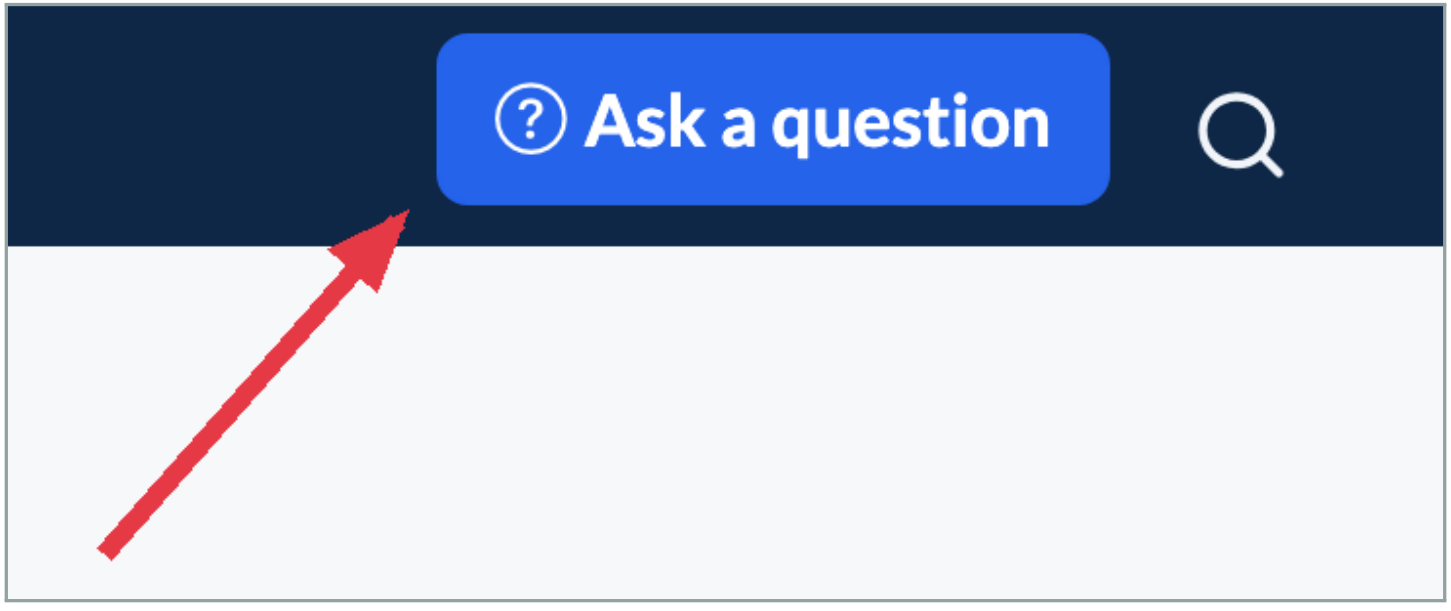
All lectures and practicals are on the course website:

ewancarr.github.io/hdr-r-november-2026

- You should have received the details by email from HDR UK.
- The site is updated throughout the course, and stays up for at least a month after the final session.
- Some sessions have a short video to watch beforehand, posted on the HDR UK Futures platform.
- Some sessions have optional homework, if you'd like more practice.

Submit your questions

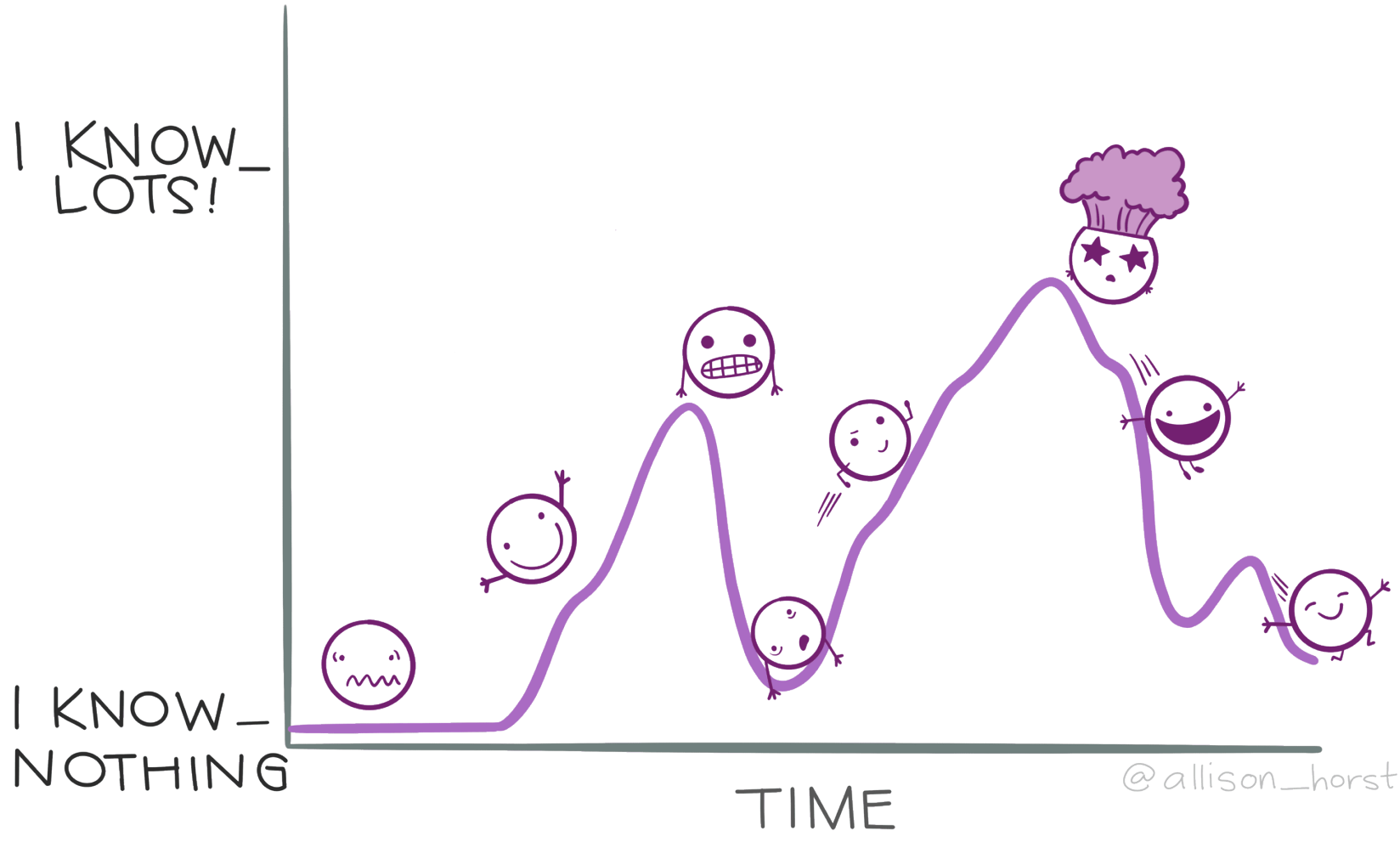
Click the Ask a question button on the course site to send me questions during the week.



Learning a new programming language is hard.

Especially when you're used to another way of working (e.g., Stata, SPSS or Python).

I tried to learn R five times before it stuck.


I KNOW_ LOTS!
I KNOW_ NOTHING
TIME
@allison_horst

Expectations

- Expect confusion. It's inevitable; you're not doing anything wrong.
- Mistakes are normal, even after many years of using R.
- Allow time; things will take (much) longer at first.
- It can help to use R for a specific project, before switching completely.
- Little and often: short, regular practice works better than occasional long sessions.

After 16 years of using R, I still:

- Regularly make mistakes
- Have to look things up and check the documentation
- Spend forever tweaking code and running endless tests, only to find the whole problem was a single comma out of place



What you'll learn

By the end of the course you'll be able to:

1. Write and run R code, and find help when you're stuck
2. Import data and transform it with `dplyr`
3. Write functions and loops to avoid repetition
4. Organise projects so they work on any computer
5. Join, reshape and summarise health data
6. Make plots with `ggplot2`
7. Write reproducible reports with Quarto
8. Work with dates, times and text
9. Know where to go next

This course is about R.

We won't cover statistical methods or modelling.
We'll cover the essential steps in every analysis:
importing, cleaning, reshaping, plotting and reporting,
and how to work efficiently in R.

"It is often said that 80% of data analysis is spent on the process of cleaning and preparing the data."[1]

But if you have questions about a particular method,
do ask. I can summarise what's available in R.

1. Wickham H (2014). Tidy Data. *Journal of Statistical Software* 59(10).

How the sessions work

A large, online class with a mix of lectures and practicals.

Lectures

- Post questions in the chat, to me or to everyone.
- Listen **or** type along, whichever works for you.

Practicals

- Post questions in the chat or put a hand up. You're welcome to unmute and share your screen.
- I'll check progress in the chat, and go through the answers at the end.

Between sessions

- Review the slides and recordings, and have a go at the homework.
- Send me questions using the form.



Every question is helpful

Someone else is almost certainly wondering the same thing.

Learning R is a journey



Different ways of working with AI

No AI

- You're working in a trusted research environment (TRE) or on a project where AI tools aren't allowed.
- Never paste patient data or output into a chatbot.

AI helps

- It explains errors, suggests fixes and answers "how do I…?"
- You write the code.

AI writes the code

- Tools like Claude Code, Codex and GitHub Copilot read your project, write code, run it and fix their own errors.
- You review what they've done.

In all cases, you need to know how the code works.

Code can run without errors and still give the wrong answer.

The analysis is your responsibility, whoever wrote the code.

You also want a say in how it's written.

Style, structure, naming, which packages to use.

Code that you (and your colleagues) can read, check and change in a year's time.

AI tends to write the most common approach. That isn't always the right one for your analysis.

You only develop this judgement by writing code yourself.

How do I use AI?

Course overview

Five sessions, Mondays 13:30–17:00.

Session	Date	Topics
1	2 Nov	**Beginner**: RStudio, data types, data frames, importing data, control flow
2	9 Nov	**Functions and data manipulation**: writing functions, `dplyr`
3	16 Nov	**Working with data**: organising projects, joins, reshaping, summarising
4	23 Nov	**Visualisation, reporting and dates**: `ggplot2`, Quarto, `lubridate`
5	30 Nov	**Advanced**: `purrr`, `across()`, regular expressions, faster code

Today's timetable

The timetable, slides and practicals for today are on the course website:

ewancarr.github.io/hdr-r-november-2026

Then click Session 1 in the sidebar.